

PyTorch vs. NCNN Runtime Comparison

Estimated ranges reconstructed from the previous Raspberry Pi analysis

Metric	PyTorch	NCNN
Pose backend	Ultralytics PyTorch	NCNN optimized runtime
YOLO pose throughput	≈ 5-8 FPS	≈ 25-30 FPS
Effective pose sampling	≈ 5-8 Hz	25 Hz target reached
30-frame buffer fill	≈ 3.8-6.0 s	≈ 1.2 s
TSSTG action inference	≈ 3-5 Hz	≈ 5 Hz target
25 Hz collection	Not sustainable	Sustainable
CPU behavior	High load / variable latency	Lower, more stable latency
Recommended use	Development / fallback	Raspberry Pi deployment

NCNN delivered roughly 4× higher pose throughput and met the 25 Hz collection target.